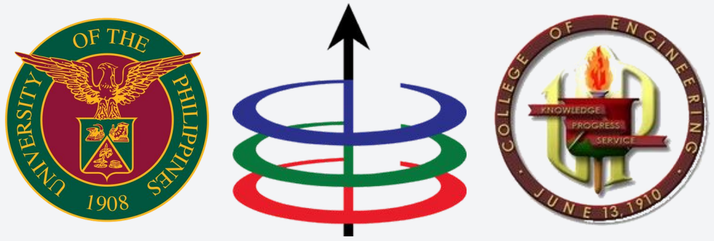

Lecture 1A
Basic C++ Programming Elements and Libraries

EEE 121 2s2526

Steven S. Sison

C++ vs Python - Hello world!

Python

```
print("Hello world!")
```

C++ vs Python - Hello world!

Python

```python
print("Hello world!")
```

C++

```cpp
#include <iostream>
using namespace std;

int main(void){
    cout << "Hello world!" << endl;
}
```

Another comparison - Solving for GCD

Suppose you want to create a program that:

- Input: u and v (where u ≥ v) are integers
- Output: ***print the greatest common divisor of the two integers***

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3      # for computing the GCD
4      while v != 0:
5          r = u % v      # compute remainder
6          u = v
7          v = r
8      return u
9
10 if __name__ == '__main__':
11     a = int(raw_input('First value: '))
12     b = int(raw_input('Second value: '))
13     print 'gcd:', gcd(a,b)
```

Recall: Euclid's Algorithm:

1. Two integers u and v where u ≥ v.
2. Let r be the remainder when u is divided by v.
3. While r is not 0:
   a. u = v
   b. v = r
4. Return u

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- Input: u and v (where u ≥ v) are integers
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24 }
```

- C++ Fundamentals
- Flow Control
- Functions
- Arrays and Strings

Before everything else...

- During our lecture sessions, use your blue book to take notes.
- Sometimes, after conducting a lecture, there will be a quiz.
- You will answer your quiz on your blue book.
- The quiz will be open notes, but you can only use the notes on your blue books.

- **C++ Fundamentals**
- Flow Control
- Functions
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Basic Syntax

```
1  #include <iostream>
2  using namespace std;
3
4  int gcd(int u, int v) {
5    /* We will use Euclid's algorithm
6       for computing the GCD */
7    int r;
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```
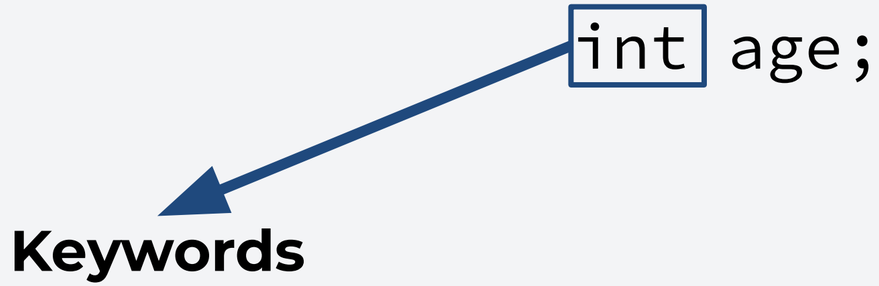

- Header files (libraries, imports)
- C++ Function
  - Basic syntax to be discussed later.
- Main Function - a C++ convention
  - This is where you put the majority of your code
- Comments
  - `//` - single-line comments
  - `/* Comment here */` - multi-line comments
- `\n` or `endl` - prints a new line
- **Always end each line with a semicolon!**

Keywords and Identifiers

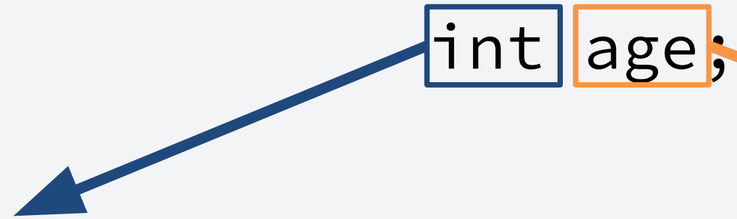
```
int age;
```

Keywords and Identifiers



- Predefined words that are used by the compiler.
- **Example**: Too many to mention, but you can find them here.

Keywords and Identifiers



Keywords

- Predefined words that are used by the compiler.
- **Example**: Too many to mention, but you can find them here.

Identifiers

- Unique names that are given to variables, classes, functions by the programmer.

Variables and Constants

- Variables are used as a storage to contain data.
  - In python, we are not required to declare variables before using it.
  - In C++, we are **required** to declare the variable first and specify its type before using it. **Syntax**: `<type> <variable_name> = <value>;`

```cpp
int q;                  // Variable declaration
int r = 0;              // Declaration with initial value
int s = 5, t = 10;      // Multiple declaration
```

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```

- We can also make the variable immutable or a **constant** by adding `const` before the datatype during declaration.

```cpp
const int u = 15;   // Constant variable u with value 15
```

Primitive Data Types

Data Type	Meaning	Size	Example Literals
`int`	integer	2/4 bytes (16/32-bit)	38
`float`	floating-point	4 bytes (32-bit)	3.14f
`double`	floating-point	8 bytes (64-bit)	3.14
`char`	single character	1 byte (8-bit)	‘e’
`wchar_t`	wide character	2 bytes (16-bit)	L‘e’
`bool`	boolean	1 byte (8-bit)	TRUE, FALSE
`void`	empty	0	None

Literals are data used to represent fixed values (but they are not constant values). **Example**: 1, 2, ‘a’, etc.

Type Modifiers

Type modifiers are used to modify the properties of primitive data types. There are four type modifiers:

Type Modifier	Meaning	Sample Usage
`short`	Used for storing small integers (-32767 to 32767)	`short a = 121`
`long`	Used for storing large integers (-2147483647 to 2147483647)	`long b = 121123138` `long double c = 0.121121L`
`signed`	Can store positive and negative integers	`signed int d = -121`
`unsigned`	Can only store non-negative integers	`unsigned int f = 121`

Data Type Conversion

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- As such, **implicit type casting** might occur during variable assignment.

```
int a = 5;          // type(a) = int
double b;           // type(b) = double
b = a;              What is the value of b?

int c;              // type(c) = int
char d = 'a';       // type(d) = char
c = d;              What happens to c?
```

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int a = 5;        // type(a) = int
double b;         // type(b) = double
b = a;            // Converts 5 to 5.00 since type(b) is a double

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What happens to c?

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c = d;            // Converts d to an integer. Question: What is the value of 'a'?
```

- However, we can also perform **explicit type casting** whenever necessary.

```
int e = 4, f = 3;
double c;
c = 4/3;                  What happens to c?
c = double(4)/3;          What happens to c?
```

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```
int e = 4, f = 3;
double c;
c = 4/3;              // c = 1.0 (4/3 = 1.33 -> 1 typecasted to double -> 1.0)
c = double(4)/3;
```

What happens to c?

Data Type Conversion

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- As such, **implicit type casting** might occur during variable assignment.

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Basic Input and Output

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- In python, we use the `print()` command to display formatted text into a screen.
- In C++, we use `cout` along with the << operator (insertion) to display an output.
  - Additionally, we can also add multiple << operator to display text.
  - We can also use the `endl` common to create a new line.

```cpp
#include <iostream>
using namespace std;

int main() {
    // prints the string enclosed in double quotes
    cout << "This is C++ Programming";
    return 0;
}
```

Basic Input and Output

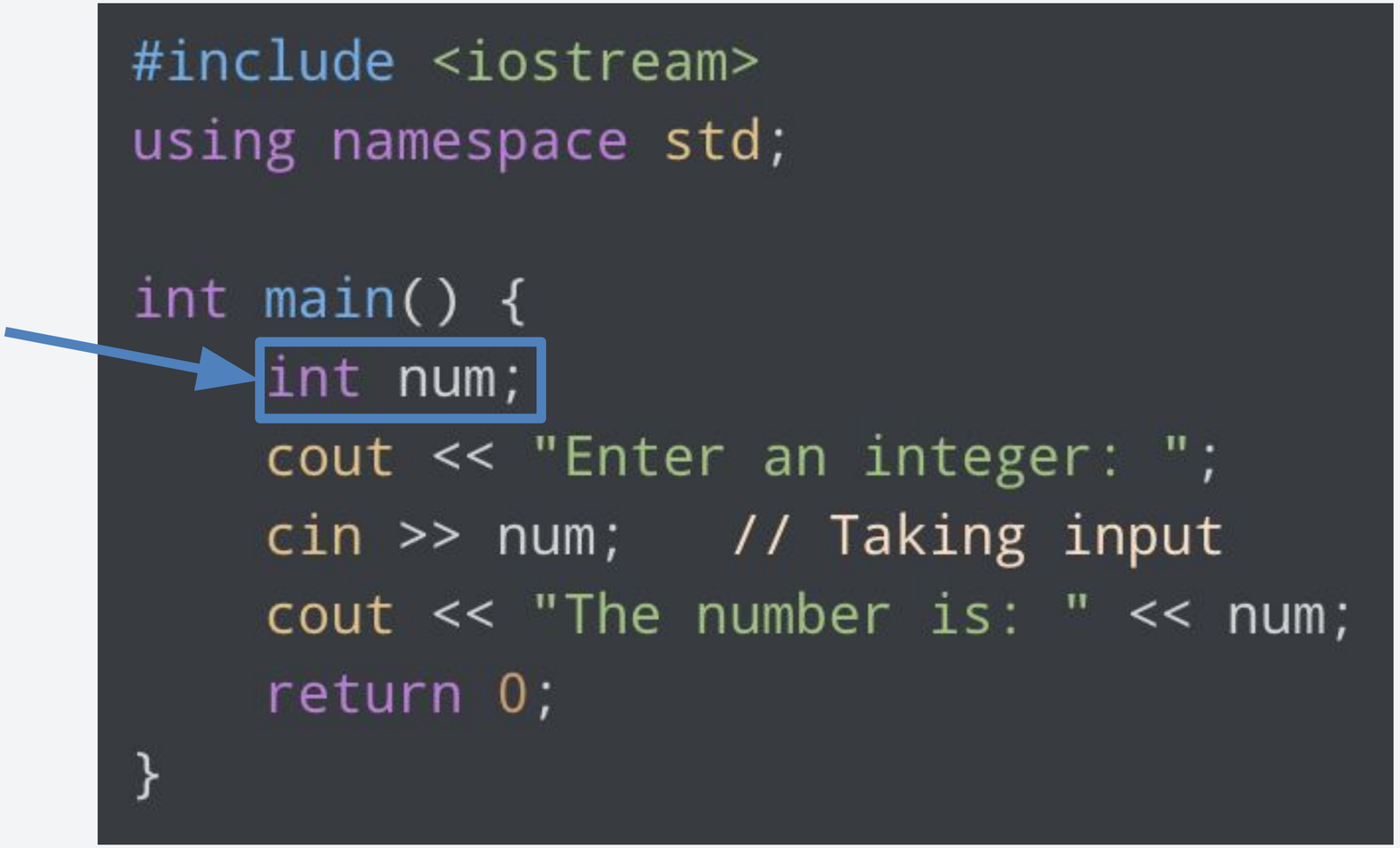
- In python, we use the `input()` command to accept user input.
  - Note that in python, the `input()` command **always** converts the input into a string.

Basic Input and Output

- In python, we use the `input()` command to accept user input.
  - Note that in python, the `input()` command **always** converts the input into a string.
- In C++, we can use the `cin` command together with the >> operator (extraction) to accept user input.
  - Note that **we are required** to initialize a variable first that will accept our input.
  - We also have to keep in mind the type of the variable that will store our input.

Basic Input and Output

1. Initialize the variable first.

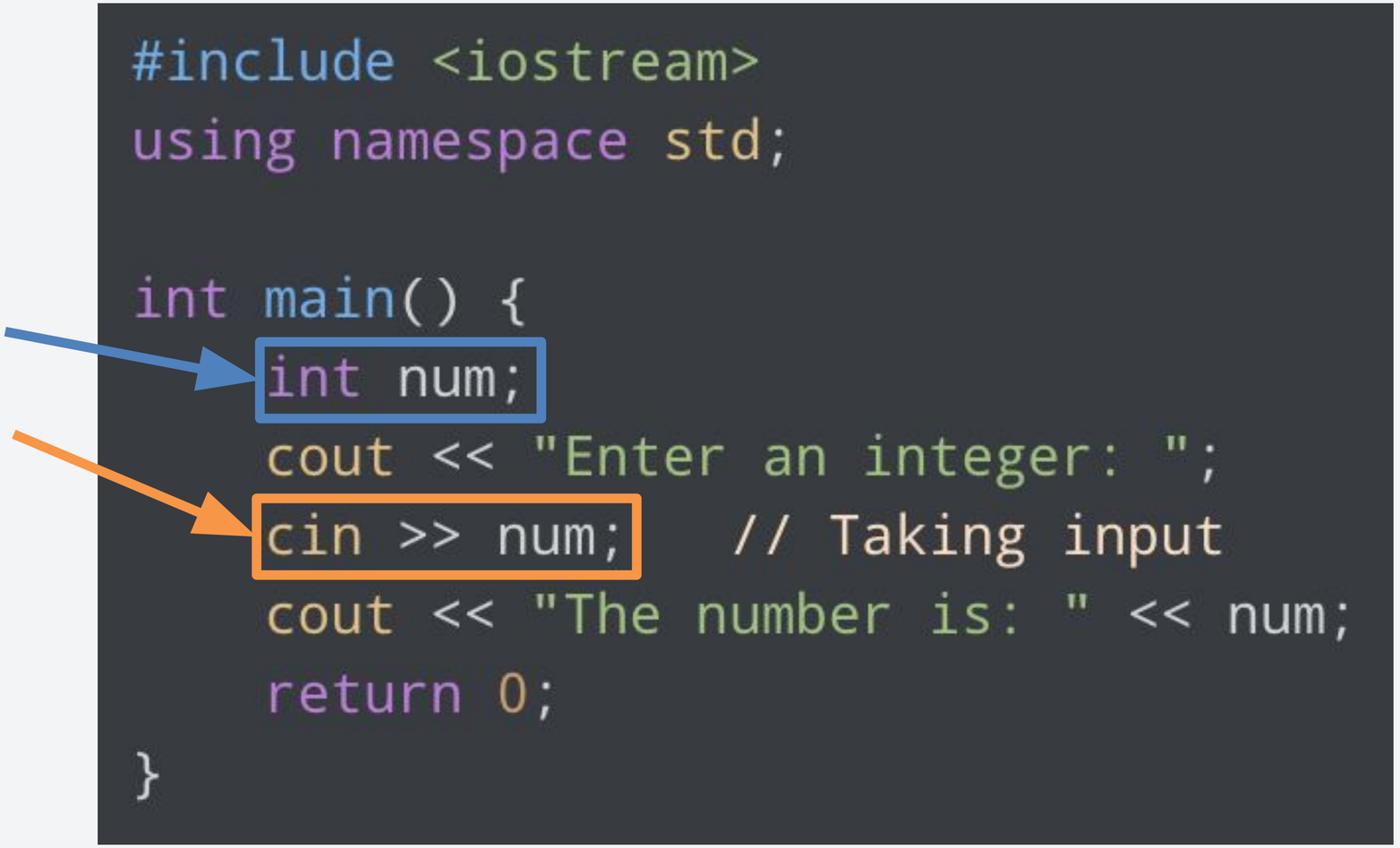


```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;    // Taking input
    cout << "The number is: " << num;
    return 0;
}
```

Basic Input and Output

1. Initialize the variable first.
2. Use `cin` to take the input.



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    return 0;
}
```

Arithmetic Operators

Operator	Description
+	Addition: Used to add two numbers or concatenate strings.
–	Subtraction: Used to subtract two numbers.
*	Multiplication: Used to multiply two numbers.
/	Division: Used to divide two numbers.
%	Modulo: Used to perform the modulo operation.

Note. See code demo.

Bitwise Operators

Operator	Description
&	Binary AND
|	Binary OR
^	Binary XOR
~	Binary one’s complement
<<	Bitwise Shift Left
>>	Bitwise Shift Right

Note. See code demo.

Assignment and Increment Operators

Operator	Example	Equivalent
=	a = 1	a = 1
+=	a += 1 or a++	a = a + 1
-=	a -= 1 or a–	a = a - 1
*=	a *= 1	a = a * 1
/=	a /= 1	a = a / 1
%=	a %= 1	a = a % 1

Note. This can also be extended to bitwise operators.

Note. See code demo.

- C++ Fundamentals
- **Flow Control**
- Functions
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Relational Operators

- These operators are used to check the relationship between operands.

Operator	Meaning	Equivalent
==	Is equal to	`121 == 123` gives **false**
!=	Is not equal	`121 != 123` gives **true**
>	Greater than	`121 > 123` gives **false**
<	Less than	`121 < 123` gives **true**
>=	Greater than or equal to	`121 >= 121` gives **true**
<=	Less than or equal to	`121 <= 121` gives **true**

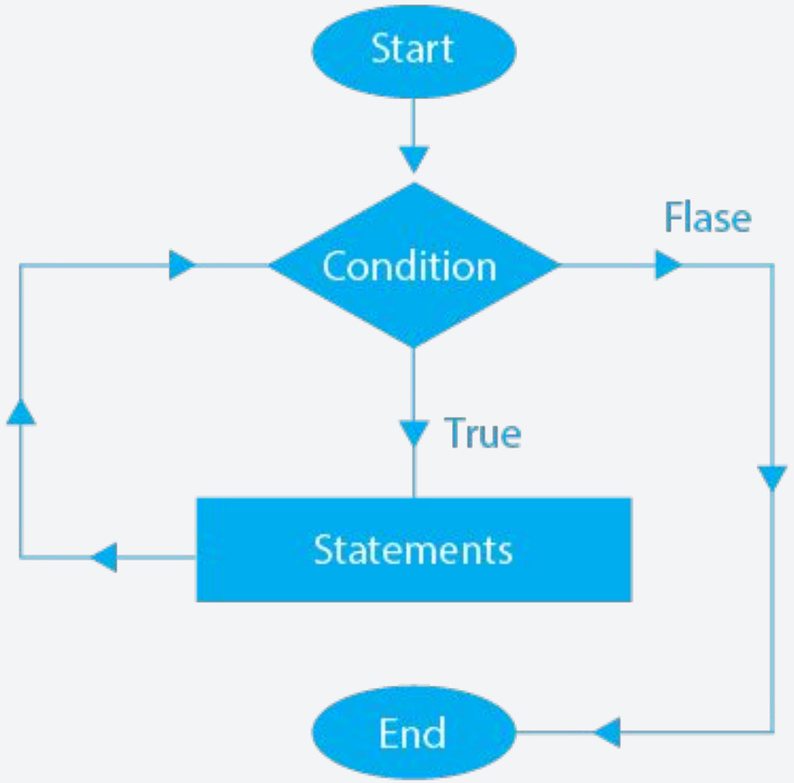
Logical Operators

- These operators are used to check whether a relationship is true or false.

Operator	Example	Equivalent
&&	expression1 && expression2	Logical AND True only if **all operands** are true
||	expression1 || expression2	Logical OR True if **at least one operand** is true
!	!expression1	Logical NOT True only if the operand is false

Looping (while, do-while, for loop)

While Loop

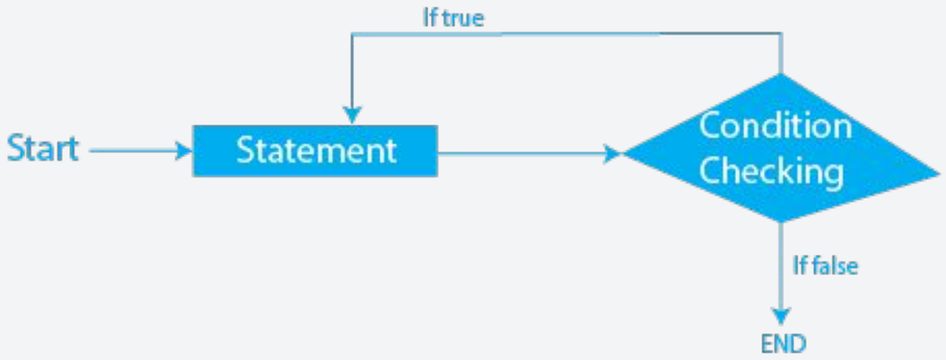



```
int i = 0;
while (i <= 100){
    i++;
}
```

While the **condition** is true, execute the **block of code**.

Looping (while, do-while, for loop)

Do-While

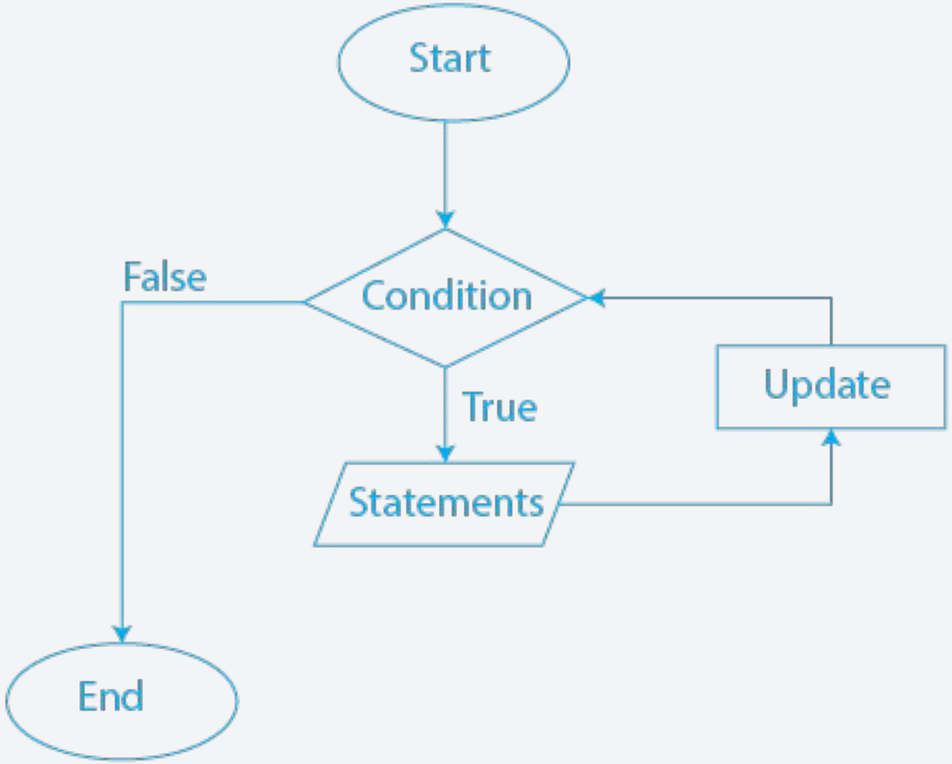



```
int i = 0;
do{
    i++;
}
while (i <= 100);
```

Execute the block of code. If the **condition** is still true, execute again.

Looping (while, do-while, for loop)

For Loop




```
for (int i = 0; i <= 100; i++) {
    // Execute code here
}
```

Do **initialization** and **check if the condition** is true. If true, **execute the code** and **update**. Repeat.

Conditionals (If-else, switch case)

If-else

```
if (statement1){
    // Execute code block 1
} elif (statement2) {
    // Execute code block 2
} else {
    // Execute code block 3
}
```

- Useful for executing code that has multiple expressions/conditions.

Switch-case

```
switch (expression){
    case constant1:
        // Execute code for constant1
        break;
    case constant2:
        // Execute code for constant2
        break;
    default:
        // Execute code if no match for any constant
}
```

- Useful for single expressions that can have multiple values.

- C++ Fundamentals
- Flow Control
- **Functions**
- Arrays and Strings

Creating and Declaring a Function

Let’s say you want to create a function that adds two numbers, you must follow the syntax:

```
returnType function_name(parameter1, parameter2){
    //  function body
}
```

```
int add(int x, int y){
    return x+y;
}
```

- The return type can be any of the primitive data types.
- In python, you don’t have to specify the data type of the parameters. In C++, you are required to specify.

Creating and Declaring a Function

- Functions should be declared before `main()`.

Method 1

```c
int add(int x, int y){
    return x+y;
}

int main(){
    int x = 1, y = 2;
    add(x,y);
}
```

Method 2

```c
int add(int x, int y);      // This is also a valid declaration

int main(){
    int x = 1, y = 2;
    add(x,y);
}

int add(int x, int y){
    return x+y;
}
```

Recall: Solving for GCD

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1  #include <iostream>
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In the highlighted code,

- What is the return type?
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- Which part is the function body?
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C++ Libraries

- We don’t have to write all the functions we need. We can reuse functions and classes from a collection called libraries
- C++ libraries are similar to Python modules
- Format:
  - Using libraries built-in with the compiler: `#include <library_name>`
  - Using own libraries: `#include “library_name” // looks at current directory`
- Some standard libraries that will be useful to us:

  - `iostream`
  - `stdlib`
  - `stdio`
  - `algorithm`
  - `vector`
  - `string`
  - `cmath`
  - `queue`
  - `stack`
  - `map`
  - `set`
  - `iomanip`
  - `ctime`
  - `utility`

Namespaces

- Provides a space to define or declare a variable, method, or classes.
- Namespaces are used to organize code in a logical manner in order to prevent conflicts between variables, functions, and objects.

```
4   namespace first{
5       void func(){
6           cout << "First Space" << endl;
7       }
8   }
9
10  namespace second{
11      void func(){
12          cout << "Second Space" << endl;
13      }
14  }
15
16  using namespace first;
17
18  int main(){
19      func();     // Outputs "First Space"
20      return 0;
21  }
```

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Notice how we also use "`using namespace std`" at the start of our C++ code? :)

- C++ Fundamentals
- Flow Control
- Functions
- **Arrays and Strings**

Array Initialization and Usage

Let’s say you want to create an integer array containing 5 elements, you need to follow the given syntax:

```
dataType array_name[size];                  // Use this if you want an empty array
dataType array_name[size] = [1,2,3,4,5] // Use this if you want to place the elements

int numbers[5] = [1,2,3,4,5]                // Sample initialized array
```

- In C++, you are required to specify the type of elements you are going to put inside the array. You can’t put more than one type.
- Arrays have **fixed size**. You can no longer modify it after initialization.
- You don’t need to know the all of the elements of the array. You can initialize it with empty elements.

Array Initialization and Usage

Array elements can be accessed through its index (**NOTE**: C++ starts at 0 index).

```
int numbers[5] = [1,2,3,4,5]                // Sample initialized array

// You can change the value of an element through the assignment operator
numbers[3] = 1       // [1,2,3,1,5]

// You can print the element of an array
cout << numbers[1]  // Output: 2

// If you exceed the array's length, an error will be raised
numbers[10]          // Array out-of-bounds
```

Array Sizes

Suppose you have an integer array, a character array, and a double array

```
int arr_1[3] = [1,2,1];
char arr_2[3] = {‘e’, ‘e’, ‘e’}
double arr_3[3] = {1.0,2.0,1.0}
```

- What is the size of `arr_1`?
- What is the size of `arr_2`?
- What is the size of `arr_3`?

Hint: size is different from length

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```

- What is the size of `arr_1`? 3*4=12 bytes
- What is the size of `arr_2`? 3*1 = 3 bytes
- What is the size of `arr_3`? 3*8 = 24 bytes

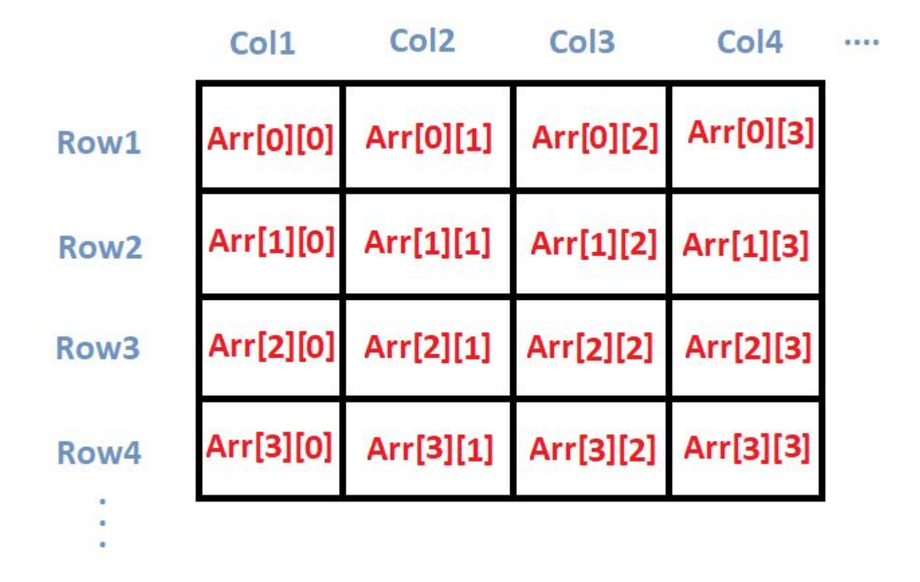
Hint: size is different from length

Recall:

Data Type	Meaning	Size
`int`	integer	2/4 bytes (16/32-bit)
`float`	floating-point	4 bytes (32-bit)
`double`	floating-point	8 bytes (64-bit)
`char`	single character	1 byte (8-bit)
`wchar_t`	wide character	2 bytes (16-bit)
`bool`	boolean	1 byte (8-bit)
`void`	empty	0

One-dimensional vs Multidimensional Arrays

- In C++ (even in python), you are allowed to create arrays with multiple dimensions. Think of it as an array inside an array.
- These are particularly useful for representing data that is inherently 2D, 3D, and more such as a matrix (2D) and tensors (3D and above).




```
int matrix[row_size][column_size];      // Initializes a 2D array

// You can acess its elements the same way you do in an array
matrix[1][2] = 2    // Sets the element at row 1, column 2 as 2
```

Strings as char array vs Strings as an Object

String as Char Array

- In C, strings are essentially just an **array of characters**.
- A string is defined as an array terminated by a null character “\0”

```
char str[4] = "C++"      // NOTE: The string requires a
                         // size of 4 because of the
                         // terminal character '\0'

// You can access its element like an array
str[2] = "C"             // str = "C+C"
```

String as an Object

- In C++, we can use the <string> library to define string as a class.
- It’s still an array of characters but using the string class provides more functionality.

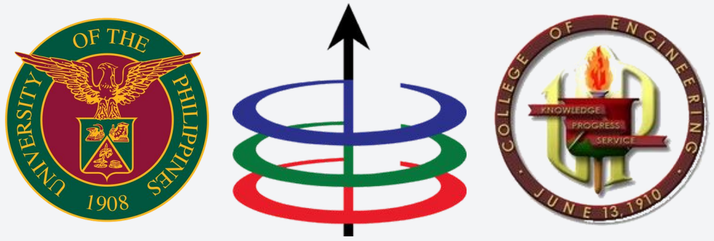
```
#include <string>

string str = "C++"

// Sample Methods
str.append(" is fun!")  // str = "C++ is fun!"
str.find("+")           // Output: 2 (It finds the first
                        // occurrence of the character
```

- C++ Fundamentals
- Flow Control
- Functions
- Arrays and Strings

Any questions?



Lecture 1A
Basic C++ Programming Elements and Libraries

EEE 121 2s2526

Steven S. Sison